7.7 DeepLab

- CNNs have shown a remarkable ability to implicitly represent scale, simply by being trained on datasets that contain objects of varying size
- still, explicitly accounting for object scale can improve the CNN's ability to successfully handle both large and small objects
- inspired by the success of the R-CNN spatial pyramid pooling method, which showed that regions of an arbitrary scale can be accurately and efficiently classified by resampling convolutional features extracted at a single scale, DeepLab implemented a variant of that scheme, which uses multiple parallel dilated (atrous) convolutional layers with different sampling rates
- the features extracted for each sampling rate are further processed in separate branches and fused to generate the final result
- the proposed "atrous spatial pyramid pooling" (ASPP) approach is illustrated in Figure 13

7.7 DeepLab

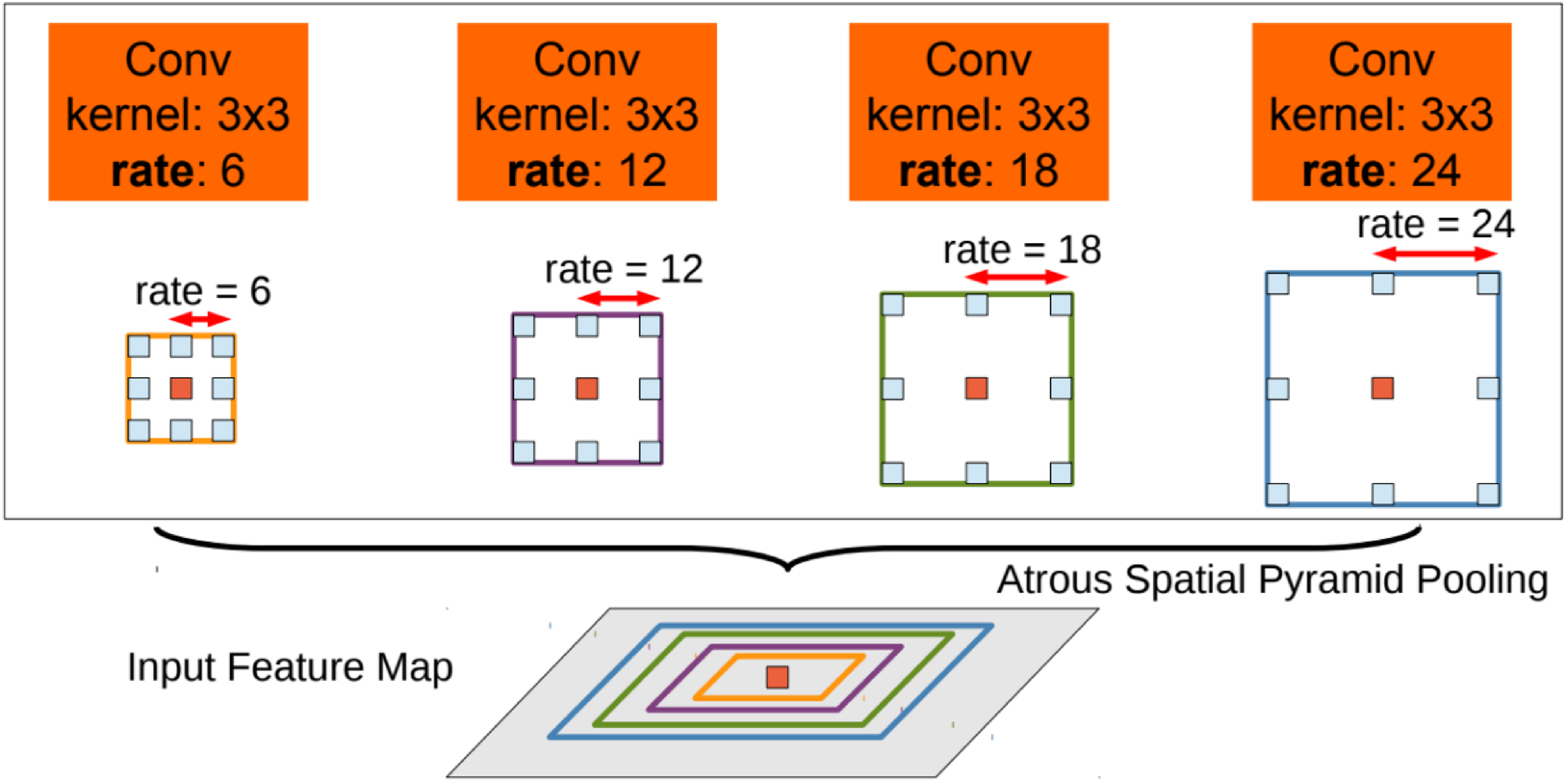


Figure 13: Atrous Spatial Pyramid Pooling (ASPP). To classify the center pixel (orange), ASPP exploits multi-scale features by employing multiple parallel filters with different rates. The effective receptive fields are shown in different colors.

7.7 DeepLab

- a high-level illustration of the DeepLab model is shown in Figure 14
- a deep convolutional neural network (VGG-16 or ResNet-101) trained in the task of image classification is re-purposed to the task of semantic segmentation by:
  1. transforming all the fully connected layers to convolutional layers (i.e., fully convolutional network)
  2. increasing feature resolution through dilated convolutional layers, allowing to compute feature maps every 8 pixels instead of every 32 pixels in the original network
- bilinear interpolation is then employed to upsample by a factor of 8 the score map to reach the original image resolution, yielding the input to a fully connected CRF that refines the segmentation results
- DeepLab can better segment objects at multiple scales via the proposed ASPP

7.7 DeepLab

- the model weights of the ImageNet-pretrained VGG-16 or ResNet-101 networks are fine-tuned to adapt them to the semantic segmentation task in a straightforward fashion
- the 1000-way ImageNet classifier in the last layer is replaced with a classifier having as many targets as the number of semantic classes of our task (including the background, if applicable)
- in order to quadruple the spatial density of computed feature responses in the VGG-16 or ResNet-101 networks, the last two pooling or convolutional layers that decrease resolution have their stride set to 1 to avoid signal decimation, and all subsequent convolutional layers are replaced with dilated convolutional layers having rate $r = 2$
- the loss function is the sum of cross-entropy terms for each spatial position in the CNN output map (subsampled by 8 compared to the original image)
- all positions and labels are equally weighted in the overall loss function (except for unlabeled pixels which are ignored)
- the targets are the ground truth labels (subsampled by 8)
- the CNN and CRF training stages are decoupled, assuming the CNN unary terms are fixed when setting the CRF parameters

7.7 DeepLab

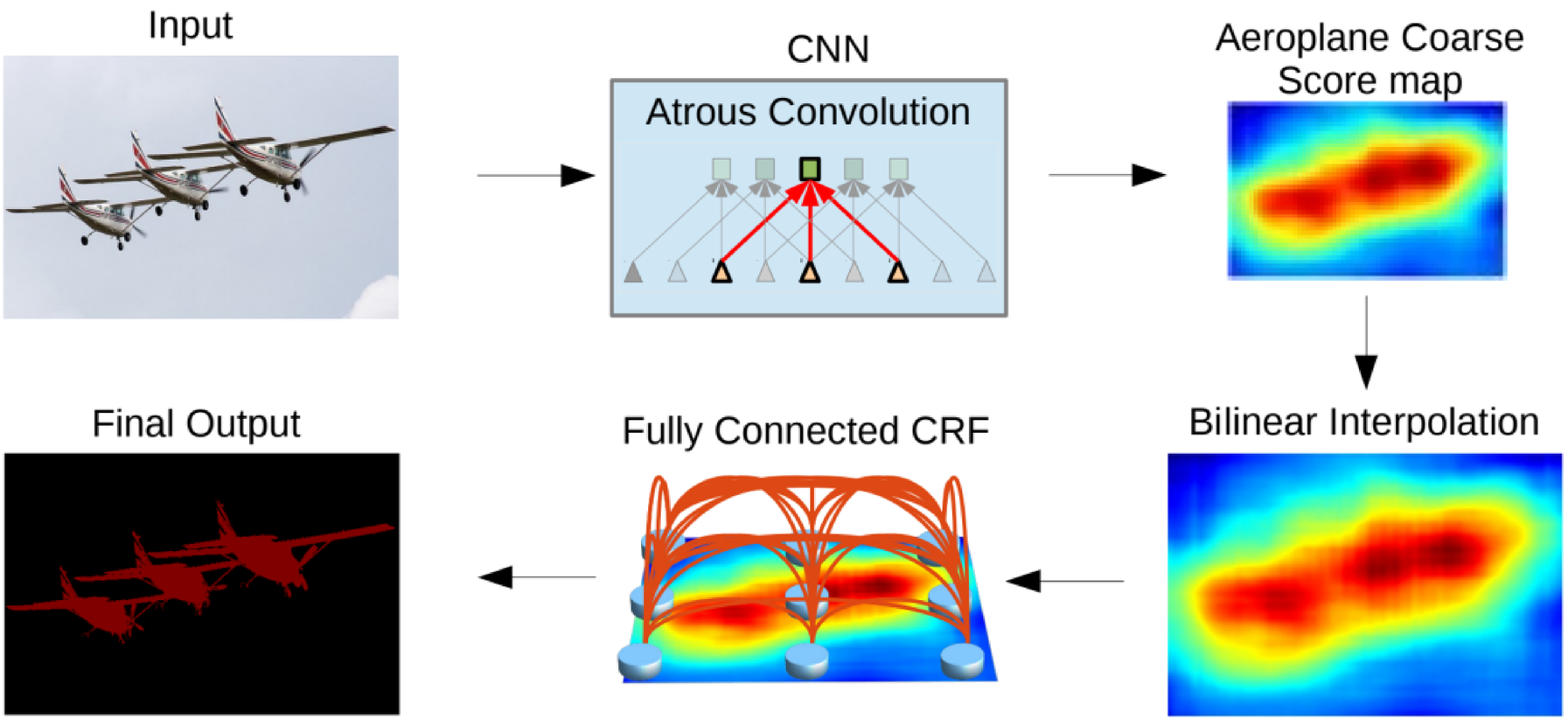
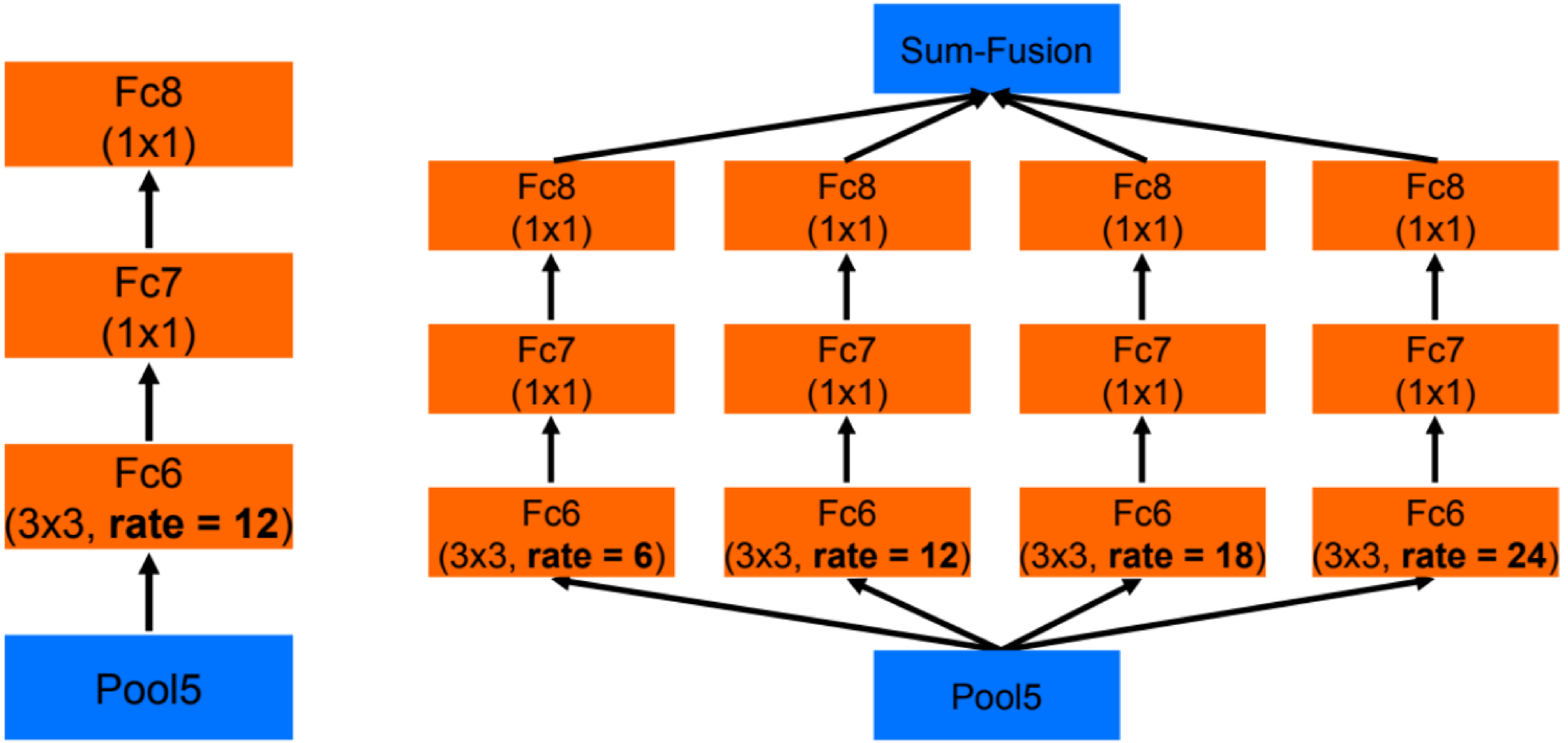


Figure 14: DeepLab model illustration. A deep convolutional neural network such as VGG-16 or ResNet-101 is employed in a fully convolutional fashion, using dilated convolutions to reduce the degree of signal downsampling (from 32$\times$ down to 8$\times$). A bilinear interpolation stage enlarges the feature maps to the original image resolution. A fully connected CRF is then applied to refine the segmentation result and better capture the object boundaries.

7.7 DeepLab

- as shown in Figure 15, ASPP for VGG-16 employs several parallel fc6-fc7-fc8 branches
- they all use 3×3 kernels, but different dilation rates r in the fc6 layer in order to capture objects of different size
- Figure 7 shows two settings:
  - the LargeFOV model, having a single branch with dilation rate $r = 12$
  - the ASPP model, with four branches and dilation rates $r = \{6, 12, 18, 24\}$

7.7 DeepLab



(a) DeepLab-LargeFOV (b) DeepLab-ASPP

Figure 15: DeepLab-ASPP employs multiple filters with different rates to capture objects and context at multiple scales.